

MODEL FAMILIES

Nine predictors · one identical protocol · one fixed seed

Classical ML

Random Forest · XGBoost · SVR (RBF)

depth-capped, L1+L2, nested inner tuning

3

models

Deep sequence

LSTM · BiLSTM · 1D-CNN · Hybrid CNN-LSTM

weights re-initialised every fold; early stopping

8.6k–77.6k

parameters

Symbolic regression

Genetic programming over the top-5 SHAP predictors

closed-form equation an officer can read and check

5

predictors

PADR — technical contribution

Phenology-Aligned Differentiable Response

agronomic constants estimated, not assumed

17

parameters